AI101 / MINI PROJECT / GROUP 50

EMPLOYEE ATTRITION

PREDICTION USING LOGISTIC REGRESSION

Using workforce data to understand patterns linked with employee exits and build a binary classification model.

HR ANALYTICS

BINARY CLASSIFICATION

PROJECT REPOSITORY

github.com/Aniket-09042008/AI101-Mini-Project-Group-50

PEOPLE ANALYTICS · MACHINE LEARNING · CLASSIFICATION

THE TEAM

Four roles. One shared project.

01

Member 1 — Aniket

Data Exploration and documentation lead

02

Member 2 — Jaswanth Adapa

Preprocessing & PPT lead

03

Member 3 — Kaushal Kejariwal

Model Training and Execution lead

04

Member 4 — Sarthak Nitin Naik

Model Analysis and Video lead

Replace placeholders with your actual names.

Can HR data help flag employees who may leave?

THE QUESTION

Why look at attrition?

- Employee exits affect workforce continuity and planning.
- HR records contain signals such as role, overtime, income and satisfaction.
- Our goal: predict whether an employee stays or leaves.

DATASET SNAPSHOT

1,470	35	2	16.12%
employee records	original columns	target classes	employees left

TARGET: ATTRITION

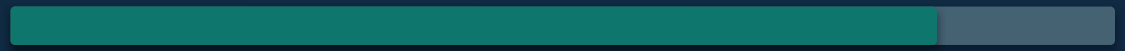
0 / STAYED

1 / LEFT

Binary outcome

CLASS DISTRIBUTION

83.88% stayed



16.12% left · 237 employees

1,233 employees stayed

Source: IBM HR Analytics Employee Attrition & Performance · Kaggle

1470 rows × 35 cols

A simple pipeline from employee records to prediction

01

INPUT DATA

CSV dataset
1,470 rows
35 columns

›

02

EXPLORE

Check data types
Missing values
Duplicates + EDA

›

03

PREPARE

Encode categories
Remove redundant fields
80/20 split + scaling

›

04

TRAIN

Logistic Regression
max_iter = 1,000
Fit on training set

›

05

PREDICT

Test-set classes
Stay (0) / Leave (1)
Probability scores

PREPROCESSING

Removed 4 non-informative columns · One-hot encoded categorical data · Omitted YearsInCurrentRole · Stratified 80% / 20% split

Train: 1,176 rows

Test: 294 rows

43 model input features after preparation

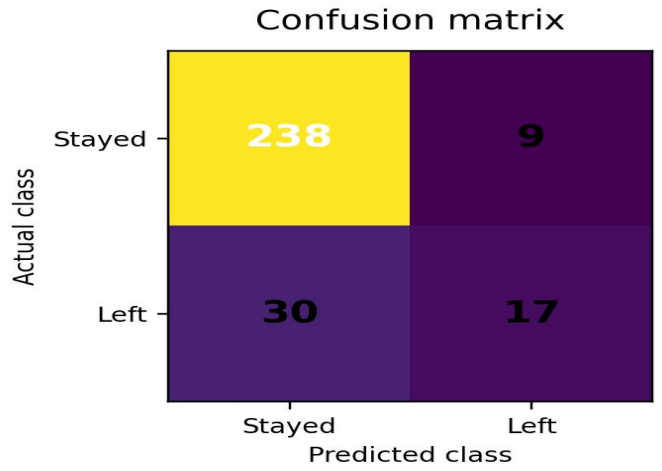
TOOLS USED

pd	**Pandas**	data tables
np	**NumPy**	numerical work
plt	**Matplotlib**	plots
sns	**Seaborn**	statistical visuals
sk	**Scikit-learn**	model + metrics

Performance on the held-out test set

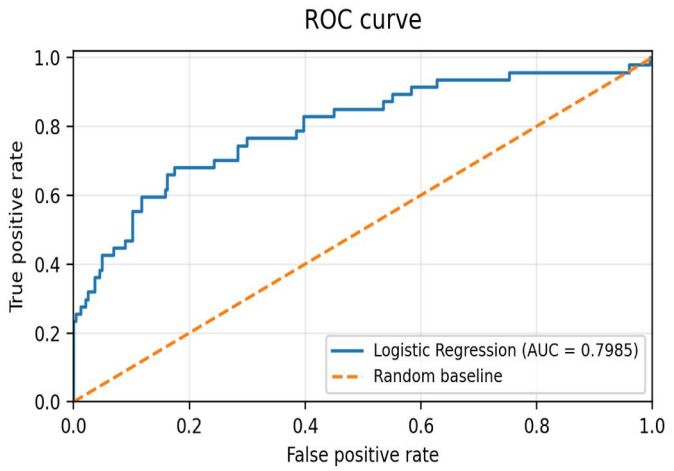
86.73%	65.38%	36.17%	46.58%	0.7985
ACCURACY	PRECISION · LEFT	RECALL · LEFT	F1 · LEFT	ROC-AUC

CONFUSION MATRIX



30 actual leavers were missed (false negatives).

ROC CURVE



AUC = 0.7985 · Above random baseline

NOTEBOOK RUN OUTPUT

```
$ python / notebook cell

Model trained successfully

Accuracy: 0.8673

ROC-AUC: 0.7985
```

Precision-recall average precision: 0.5735

Interpretation: accuracy is strong, but recall for the minority “Left” class is lower — class imbalance remains important.

What worked, what needs attention, and who contributed

WHAT WORKED

- A clear preprocessing workflow made the dataset ready for modelling.
- ROC and precision-recall metrics added context beyond accuracy.

WHAT NEEDS ATTENTION

- Only 36.17% of employees who left were detected at the default threshold.
- Future work: test class weights and decision thresholds to improve recall.

TEAM CONTRIBUTIONS

01 **MEMBER 1**

Data Exploration and documentation lead

Name : Aniket

02 **MEMBER 2**

Preprocessing & PPT lead

Name: Jaswanth Adapa

03 **MEMBER 3**

Model Training and Execution lead

Name: Kaushal Kejariwal

04 **MEMBER 4**

Model Analysis and Video lead

Name: Sarthak Nitin Naik

AI-USE DISCLOSURE

ChatGPT and Claude were used to support coding, clarify machine learning concepts debug errors, and organize the project documentation and presentation